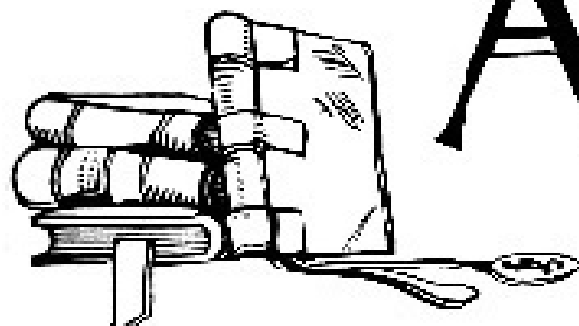


An illustration of a stack of books, with a dental mirror and a dental probe resting on top of them.

THE ARCHIVES OF DENTISTRY

SUCCESSOR TO
Missouri Dental Journal, also Consolidated
with
New England Journal of Dentistry.

Vol. VII. No.12.]

DECEMBER 1890.

[NEW SERIES

An illustration of a stack of books, with a dental chair and dental tools (forceps and a mirror) positioned next to them.

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Title: The Archives of Dentistry, Vol. VII, No. 12, December 1890

Author: Various

Editor: W. H. Eames
C. S. Stockton

Release date: January 8, 2020 [eBook #61132]
Most recently updated: October 17, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/61132

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OF DENTISTRY, VOL. VII, NO. 12, DECEMBER 1890 ***

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VOL. VII., No. 12.]

DECEMBER, 1890.

[NEW SERIES.

Original Articles.

Original Articles.

No article will be published in this department that has been read before any society or has appeared in any publication.

The editor does not hold himself responsible, in any sense, for the views expressed by the authors of original articles.

Any article intended for this department should be received by the first of the month previous to its publication.—Ed.

SOME OF THE ABUSES OF CROWN AND BRIDGE WORK.

BY WALTER M. BARTLETT, D.D.S., ST. LOUIS, MO.

There are many points in crown and bridge work requiring particular attention which many operators seem entirely to overlook in the construction of this special class of work. It might be of interest to you to turn your attention in this direction.

In the first place, the use of crowns has been abused to a great extent by the crowning of all class of roots, some of which are only fit to be crowned by a good, strong pair of forceps. These roots may be divided into three classes: first, those that have been in a diseased condition for years and have been a source of constant irritation to their owners; second, those that are only held in by their gum attachments; third, those whose walls have been fractured, said fracture only extending down as far as the process.

As the result of many inquiries in regard to the class first mentioned, I find that the majority of operators after spending weeks and even months in careful preparatory treatment, have met with very little success in crowning such cases. They invariably cause some annoyance to the patient. In the majority of cases the trouble is that of inflammation of the surrounding tissues, caused by the constant moving of the root during the process of mastication. After a root has been in a diseased condition for a year or

more, it is questionable whether treatment, however carefully given, will place that root in a condition to be crowned and do the work of the original tooth.

The second class is beyond the bounds of practical dentistry, and no operator who has any regard for his reputation will attempt this class of crown work, unless he has a mercenary object in view.

All who have attempted the third class have made many failures. When there is a fracture extending from the centre of the root, running down to the process border in a slant, no crown can with any satisfaction be properly adjusted, owing to this movable portion of the root. Were this portion removed, the root would be placed in a position impossible to crown. Then again, in case the crown is successfully placed, how long will it last? Probably not more than a year. There is bound to be a decaying of the root at the site of the fracture. From decay, that fractured piece will soon become loosened, and the chances are that it will in some way work its way out, and as a result, there is a cup to accumulate food, which will cause, later on, a very offensive odor. A case of this kind cannot be ferrelled without the results already mentioned. Should the root be split down through its centre, then a crown could be used without fear of any bad results, by bracing the root with a strong ferrell made of platinum, letting it extend down to the process border, thus avoiding the possibility of any secretions getting in between the fractured parts. The difference between the two cases can very readily be seen. In one case, there is a loose, disconnected piece of tooth structure only held in place by the membrane surrounding it. In the other case, there are two firm parts which are firmly held in place by their bony socket.

These three classes of roots are being crowned day after day by men who are considered competent practitioners, and their only success is the satisfaction of knowing that they retained the root, or roots, for six months or a year's time.

These are only some of the abuses of crown work, to say naught of teeth which could be properly filled, that are sacrificed for crowns which can never take the place of teeth properly filled.

Bridge work has taken the place of gold plates to a certain extent. Still, however good a thing it may be, it also is greatly abused. In my mind, a

bridge is not a practical dentiture when it has to span the entire arch supported only by three or four roots, or, say teeth. It has no superiority over a gold plate in such a case, because the latter can be kept clean, while the former can not, and the best of bridges must be removed at times to be reset. The chief objection to a full case of bridge is its becoming loose, which is no doubt caused by the unbalanced movement of the roots, as pressure or weight cannot be brought to bear upon three spans surmounting four pieces at the same time; one piece will surely give way to the undue strain.

The abusive use of open-face crowns must not be overlooked. These classes of crowns for bridge work should never be used, as they are simply not practical. We have yet to see one which will last any length of time. They have no strength nor durability. One of their weak points is the band around the labial surface of the tooth which in course of time cracks. Another is the free access which the secretions have to do their work upon the cement which holds the crown in place.

We are taught that a cement filling is not a permanent one, and especially should not be used below the gum border; but we are advised to use open-face crowns and to set them on with cement. If cement will not last when put into a cavity of a tooth at the gum line, we should hardly imagine that it would last very long when only protected by a thin gold band placed below the gum line, where secretions have continual play at it.

Another class of bridge work which should be done away with is what is called pin bridges, where places are bridged over, getting the necessary support by drilling cavities into sound teeth to insert the pins necessary to support the bridge. This is a class of work which simply will not last, owing to the lack of proper strength required to support a dentiture of that sort. If a good tooth must be sacrificed, why not crown the necessary teeth and get something from the start that will be of lasting service?

In my mind it is far better to have a bridge rest fully upon the gum instead of having a shelf-like projection over the ridge. It is said that the advantage gained from this class is its cleanliness, as there is no way for the food to become lodged during the process of mastication, as is the case with the shelf-projecting bridge. One would naturally imagine that food could find

its way underneath, between the gum and the teeth, but strange to say, there is very little that ever finds its way to those places.

In closing, I will offer a suggestion which may, or may not, be new to you. It is this:

For investing material, I recommend common moulding sand, in proportion of one-half plaster and one-half sand. After allowing the mass to set hard, trim it down to suit the case invested, and then with a hair-brush paint the upper surface with a solution of borax water. The advantage of the use of sand as an investing material is that from the change of the color of the mass from a deep yellow to a lighter shade you can perceive the exact moment at which your case is thoroughly dried out. The borax has the property of preventing the mass from splitting. Its greatest qualities are its strength and non-contractible properties, it having an advantage in this respect to pumice, asbestos and other materials generally used for this class of work.



ALUMINUM.—A new process of extracting this metal has been discovered, it is claimed, by which the metal can be produced for a few cents a pound. Prof. Joseph M. Hirsh, Chicago, Ills., is the one whom it is claimed has solved the problem of extracting the metal from an aqueous solution.

Societies.

Societies.

OHIO STATE DENTAL SOCIETY.

The Ohio State Dental Society held its sixth annual session in the Senate Chamber, Columbus, O., Oct. 28th, 29th and 30th, 1890, and was opened with prayer by Dr. Gladden.

Dr. W. H. Sedgwick, of Granville, O., delivered an address.

Fifty years ago the extraction of teeth was largely practiced. A man's ability was gauged by his mechanical skill. Since then there has been quite a change which has been wrought by the agency of intercourse, societies and literature. He compared the cheap dentist to the physician who would advertise, best prescription, fifty cents; medium quality, fifteen cents; a lawyer who had best opinions, \$9.00; poor ones, 5.00; or a minister who delivers his sermons in job-lot style and attends Sunday-school in the bargain.

Dental students should spend two or more years in office pupilage before entering college and they should not be received at all unless they agree to spend their time and attend the required college course before practicing. The college should reciprocate and not accept a student who has not had this previous pupilage.

Every practitioner should attend associations, and these associations should be enlivened by clinics. All should take part, and the younger members especially. We need a dental law in Ohio, and the time has come to do something. It should be made unlawful to extract teeth which can be saved. The dental profession does not investigate, read and study enough. A post-graduate course would be a good thing for all.

DISCUSSION OF PRESIDENT'S ADDRESS.

DR. H. A. SMITH:—Drs. Harris and Taylor were the first to inaugurate dental schools. Classical education will in time be one of the requisites for admission to dental colleges. The three years' course is the cause of so many students this year.

DR. M. H. FLETCHER:—Careful observers produce the best dental literature.

DR. G. H. WILSON:—Our colleges are coming to require better previous education. A special course for digital culture would be a good thing. Our excellence in the latter accounts for our supremacy over German dentists. They are all brains.

DR. J. TAFT:—Schools for instruction in the different branches, and graded schools would be a good thing. The class of students to-day is better than five or ten years ago.

DR. C. R. BUTLER:—The demand of to-day is for better students and better dentistry.

DR. D. R. JENNINGS:—The trouble with office pupilage is that the student is only taught how to master the broom, coal scuttle, dirty flasks, &c.

DR. J. TAFT:—A student should be a man of good breeding. It has been urged that he should be examined by an oculist before commencing.

DR. F. SAGE:—Young members should take part in societies; arrangements should be made some time before, subjects assigned and persons appointed to open the discussion of the same. Dental students need habits of study, so they can grasp everything taught at the college. Dental teaching is more to be remembered than almost any other—not any should be lost.

Dr. Frank Hamilton of Columbus, O., lectured upon "Surgery of the Mouth and Jaws," enlivening the same by presenting cases. He said tumors were the most frequent cause; necessitating the removal of a jaw or a portion of it. When the superior jaw is to be removed, a flap is dissected back, commencing at the center of the lip and carrying the knife around the nose and up through the wrinkle under the eye. This leaves little disfiguration. One or two incisor teeth are next extracted and the nasal and molar processes are severed with the saw and the whole dislodged with strong

pliers. Bleeding arteries are caught, and it is sometimes best to ligate the carotid. Bichloride solution is used throughout the operation.

Discussion—DR. C. R. BUTLER:—There will be little disfiguration if the knife follows the natural depressions in cutting the flap.

Dr. J. Taft read a paper by Dr. N. S. Hoff, of Ann Arbor, Mich., "Your old men shall dream dreams; your young men shall see visions."

One person suggests a change in our laws; one that we have a home on a high mountain in Georgia; another that we have a National Board of Dental Examiners and a higher degree given than D.D.S., and another has offered a dental society aid towards securing it a home. The author suggests that the Ohio State Society, which has been wandering around, have a home, where its sessions could be held, and connected with it a library, a museum and facilities for clinics and lectures. Columbus, the capitol, would be a good place, being central and having the usual facilities of a large city. The local and district societies could be auxiliary to it. A fund, a room or building secured with some one in charge. All the specimens, appliances, &c., contributed by members. Specialists should be secured to lecture at the meeting or other times, and along with this, a dental journal devoted to its interests.

Discussion—DR. J. TAFT—The suggestions are not impossible and can be carried out if all would take part and aid with money and all the specimens and appliances they can spare.

Dr. Frank Sage, Cincinnati, O., next read a paper entitled, "Intimate Diagnosis of Lesions Effecting the Teeth."

Idiosyncracies of persons have much to do with lesions of the teeth, but there are often other causes also. A patient comes from some other operator, and ignorant of what has been done, he is apt to be misled. All causes which are not directly associated with the trouble, if first eliminated, the diagnosis is much easier. The prejudice and ignorance of the patient must be considered. A previous condition should be studied, and its influence at the present time weighed that we be not led to error. We must not be influenced by mere probabilities. Teeth once affected are quite likely to be mistaken as the cause of trouble again.

Discussion—DR. OTTO ARNOLD:—Instinct is a pretty good guide in difficult diagnosis.

DR. R. D. JENNINGS:—Physicians know very little about the teeth.

DR. M. H. FLETCHER:—Pulp stones are often the cause of lesions very difficult of diagnosis.

Dr. L. E. Custer, of Dayton, O., read a paper upon "Chloride of Methyl."

Since Dr. B. A. R. Ottolengui introduced the ether spray for obtunding sensitive dentine there have been a number of such agents introduced for the same purpose.

There has been considerable confusion regarding the action of volatile agents for sensitive dentine. Dr. Ottolengui in previous paragraph first used ether, supposing it to dehydrate further than alcohol or hot air; but it does not, because it has no affinity for water and because it produces cold, which opposes vaporation of the dentine's moisture. Dr. Curtiss in using nitrous oxide is led to the belief that it obtunds because it dehydrates. But that is a mistake, for the same reason as for ether. When these agents obtund sensitive dentine, it is because they reduce the temperature far below normal, and the crown of the tooth being easily isolated, it is not difficult to reduce the temperature so low as to render it entirely void of sensibility.

Chloride of methyl being an ether quite low in the series, volatilizes at 74° Fahr. between ether and rhigolene, and is capable of reducing the temperature to 40° below zero, if necessary. It volatilizes with considerable rapidity with the warmth of the hand, and so does away with any blast apparatus. Is not as inflammable as ether. Is not a solvent of caoutchouc, and so it may be conducted through a rubber tube to the tooth and its flow regulated by a thumb-screw. It volatilizes quicker than ether, so that the surrounding parts are not saturated.

Its objections are that it is a general anæsthetic and at the present time rather costly. It is all that Dr. Rhein claims for it, and is to be preferred as a cold-producing agent for sensitive dentine.

But the more experience we have the better we can control our patients by a sort of personal magnetism, and dehydration will answer our purpose without resorting to such dangerous methods as the reduction of temperature.

Discussion—DR. C. R. BUTLER:—General anæsthesia is probably slightly produced, which accounts for part of the effect.

DR. J. R. CALLAHAN:—A sort of hypnotism is produced when obtunding is suggested by the use of alcohol or hot chloroform. For the worst cases, a little nitrous oxide is effective.

DR. OTTO ARNOLD:—If you will impress the patients that everything is being done to ease the pain, you will succeed. The Niles apparatus, which throws a jet of vaporizing alcohol upon the dentine, is very effective. A few inhalations of chloroform have a good effect.

DR. C. R. BUTLER:—Small amounts of chloroform are very dangerous. It would be as safe as nitrous oxide if the people had as much confidence in it.

DR. H. A. SMITH:—Doctors disagree as to that. Dr. Hamilton says "ether may be used to the 'ether glow' without a particle of danger." Quacks are often successful, because they give a full dose. The use of such anæsthetics is dangerous, but quacks, like fools, rush in where wise men fear to tread.

DR. P. S. BOLLINGER:—I have tried nearly all obtundents, but have been most successful in working upon the imagination of the patient. Purchase a firm hold on the instrument.

DR. C. R. BUTLER:—The fountain-pen answers beautifully as an instrument for using volatile agents.

DR. J. TAFT:—The rubber cloth must be used. There should be dehydration and reduction of temperature. Cold is so successful as an obtundent, because it does not vary like other agents. A combination of therapeutic agents might be used in the Niles atomizer.

Dr. C. R. Butler, Cleveland, O., Paper, "A Means of Holding the Rubber Dam, While Operating Upon Labial Surface Cavities."

Many appliances for this purpose have been devised, but when applied are often very painful and difficult to carry the rubber over.

In these very low cavities a hole is drilled below the margin of the cavity far enough to allow it to be excavated and still leave enough for strength, and into this hole a platina-iridium or gold wire is screwed, first dipping in phosphate cream. The cud is cut off within an eighth of an inch of the

cementum and the rubber carried over the projecting point, by which it is afterwards held. Sometimes two such pins will be found necessary.

In performing this part of the work never cut a thread in the cementum. If the shouldered spear-pointed drill is nicely gauged to the wire, like the S. S. White retaining screw-set, the wire will cut its own way in.

When the filling proper is completed, dress off the wire pins as though they were fillings.

Discussion—DR. J. T. TAFT:—The Palmer clamps have been a source of pleasure in my hands. They fit each tooth so nicely that great stiffness is not required.

DR. H. A. SMITH:—If a chisel is rested against the cementum and lightly tapped, the assistant can easily hold it. Bleeding gums cause much time to be lost.

DR. F. SAGE:—Sharpen the large end of a steel pen and drive it in the cementum. This has the general form of the festoon of the gum.

DR. M. H. FLETCHER:—Spunk and dry plaster of paris used when the rubber leaks, takes up moisture and sets, adhering to the teeth and rubber.

DR. CHAS. MILES:—Thrust a heated instrument through the rubber and it will adhere.

DR. J. R. CALLAHAN:—The How clamp is a good appliance for this class of cavities.

DR. C. A. BUTLER closed by saying there was no cut in the cementum by using the screws. They are dressed like gold fillings.

Dr. J. R. Callahan, Cincinnati, O., Paper on "Hypnotism."

The old terms, animal magnetism, clairvoyance and mesmerism, have gradually come to be called hypnotism. They were used principally by charlatans, and often for wrong purposes, and so came into disrepute. It was supposed that the operator had some supernatural power.

Prof. Bernheim says hypnotism is produced by what he calls suggestion. The operator says certain things, and the patient imagines them to be true. During this time all the functions are normal, except the first time, when

there is usually more or less excitement. The proportion of persons susceptible to hypnotism is placed at about 18 per cent. in both sexes.

This has been used with considerable success in organic lesions, hysteric, neuralgic and rheumatic affections. It is difficult to produce when an operation is in view.

There is still considerable superstition and ignorance regarding hypnotism, and it is yet a field for study. A professional hypnotist is called an infidel, a crank and all such bad names. Its abuse has brought it into disfavor. Every time it is used the patient becomes more susceptible to its influence. Hypnotism should not be used in dental practice until it has a better name and more is known about it, but it would be quite in place at a clinic.

Discussion—DR. E. G. BETTY:—Dr. Dunn says hypnotism is a physiological process, but that it may be successful, the mind must be calm.

DR. D. R. JENNINGS:—Here is an article from a paper on hypnotism, which I will read. The author holds that strong-willed persons become self-hypnotized when they will do a thing. It becomes almost impossible for them to break away from its influence. When the time comes, they do it as though some blind force was controlling them.

DR. JENNINGS could see no difference between mesmerism and hypnotism. Christian Science is another of the same kind and just as effective.

DR. H. A. SMITH:—Hypnotism should not yet be used in dental practice. It is dangerous.

DR. W. H. WHITSLAR:—Hypnotism is a sleep and is not induced by supernatural powers. The patient becomes afraid of the operator after its use. The Russian and French governments have passed laws regulating the practice of hypnotism.

It should not be used in the dental office at this time.

Dr. Callahan replied that the power of hypnotism was in the patient and not the operator. The operator suggests and the patient imagines. Its practice is dangerous and should be regulated by law.

Dr. F. Jacobs, of Newark, O., explained his method of making sections of teeth for the microscope. After cutting from both sides to the centre upon a coarse wheel, the thin section is placed upon a cork and held against a fine

wheel kept wet all the while. It is stained with carmine ink or sassafras oil and mounted in balsam.

Dr. Fletcher, Cincinnati, O., Paper, "Dental Implantation."

We have been relying upon the wrong tissue for success. The pericementum was formerly supposed to be necessary for success. Dr. G. L. Curtiss was first to remove this portion before implanting. Very few cases are permanent successes. Reports of failure are now coming in quite frequently. Dr. Fletcher used for experiments a goat and the teeth of a small dog. Owing to difficulties in holding the teeth in position, they were maintained in place by sewing over the integument. The teeth were inserted in the bone rather than the alveolus. Parts of the crowns were removed so as not to prove so great an irritant to the integumentary covering. Bichlorid was used throughout.

After nine months they were opened into. In all cases periosteum had grown over them. All but two had almost entirely disappeared. In one case, cementum became reorganized and grew at the expense of the dentine. The pericementum cuts no figure in implantation of teeth.

Implanted teeth can never be successful for these reasons: First, teeth are developed from a different structure than bone. If dried pericementum took on new life and performed its functions as before, it would be a success. If we could implant immediately after a tooth is extracted, success is more probable. It would be equal to skin or bone graft. If the cementum becomes reorganized, it can grow only at the expense of the dentine. If resorption takes place in vital tissues it certainly is so in dead tissues. So it would be but a few years until the whole root would be replaced by a new material, probably bone, since it is produced by bone-producing tissue.

The experiments of Drs. Hopkins and Penrose were reviewed in this connection and the following deductions drawn. First, sterilized bone in favorable conditions undergoes organization. When acted upon by periosteum is absorbed, and when in a narrow cavity, and not too large in bulk, organization and absorption both take place. Second, these processes go on most actively between five and eight weeks.

According to Dr. Wm. Savory, tight ivory pegs were more quickly absorbed than loose ones. He draws the following conclusions:

First—the operation of implantation has fallen into disrepute, either from failure or loss of confidence.

Second—when the root is covered with dried periosteum the membrane is absorbed before union takes place, and when union does occur it is probably that of ankylosis between vascular cementum and the surrounding bone. In view of which, teeth with much thickness of cementum are to be preferred, and should be denuded of the membrane before the operation.

Third—organization takes place better when the tooth is moderately tightly fitted in solid bone, and for this reason the cicatrix of bone formed after the extraction of a tooth is the most favorable place for implanting teeth.

Fourth—that the resorbtion and rebuilding of the tissues of the body necessitate the absorbtion of the dentine of the roots of implanted teeth, and thereby their loss. But that as a temporary replacement of lost teeth, the operation of implantation is justifiable to those who comprehend it to be such.

Discussion—DR. BUTLER:—I have not performed this operation, as I was doubtful of its final success. The question now is whether it is justified by its duration of three to five years, and the patient's liability to disease, etc.

DR. SAGE:—I wish to know what was considered large or small ivory peg.

DR. FLETCHER:—The experiments of the men go to show that bone will become organized when ivory will not. The only part of a tooth that may be organized is the cementum. The line of reorganization or absorbtion is a softened tissue, so when the process reaches the crown and is about complete, the crown drops off. If a very fresh tooth is implanted, the membrane may be accepted.

The following officers were elected for the ensuing year:

President, E. G. Betty, Cincinnati, O.; First Vice-President, J. R. Callahan, Cincinnati, O.; Second Vice-President, G. H. Wilson, Painsville, O.; Secretary, Otto Arnold, Columbus, O.; Assistant Secretary, Henry Barnes, Cleveland, O.; Treasurer, C. Q. Keeley, Hamilton, O.

H. T. Smith re-elected member Board of Examiners.

DISCUSSION OF DR. BLAISDELL'S PAPER. [\[1\]](#)

DR. SMITH:—Mr. President, the essayist this evening has presented a paper which is a resume of the methods of practice of many practitioners throughout the country, but although they have given their methods of capping exposed pulps, I do not see that they give their results, and he cites but one case and pronounces that a failure.

The most important thing to avoid in the treatment of exposed pulps, as held by the authorities, is pressure, and I endorse that opinion most thoroughly. I believe a large proportion of failures in the capping of exposed pulps is due to the lack of skill on the part of the manipulator. We speak of accidental exposures in excavating; I believe that that exposure is almost inexcusable. I don't believe any graduate of the dental school of Harvard College has any right to punch an excavator into a pulp unless he intends to do it when he starts. It may be a little personal to say that my experience in accidental exposures is very small, but such is the fact. I have had cases where perhaps the exposure was excusable. In cases of a malformation of the pulp or tooth, but under ordinary circumstances, in cases that we usually meet, such exposure of the pulp I believe to be wholly inexcusable. The practitioner of to-day in opening into a cavity knows, or should know, that he is liable to come into contact with the pulp, and he therefore ought to excavate with the greatest possible care, and with the careful instructions given in the schools to-day there is no excuse for his exposing that pulp.

The method cited of using gutta-percha, I do not believe in, and the placing of gutta-percha directly on the pulp, I condemn, not from my own experience, but from my observations of cases treated by other practitioners, the patients have come into my hands afterwards. Trouble has resulted in every case that I have seen, but they may have been cases which should not have been capped, and perhaps it is hardly just to deny that some were successes.

The pulp, while it is an organ of great sensitiveness and extreme delicacy of structure, I believe to be most persistent in its vitality. The capillary circulation of the pulp, as you all know, arising from the vertical vessels forming loops, prevents a combination near the surface, and the absence of lymphatics in the pulp prevents medicaments from doing much good, as

would be the case upon serous membrane. Where the trouble is diffused through the entire membrane, in many cases the application of arsenious acid will only destroy a part of the pulp, and has to be re-applied again and again.

If a tooth has been aching, and there is congestion about the pulp, or an exuding of pus, I invariably destroy it. I don't believe a pulp can be brought to a healthy condition to stand capping after it has reached that stage. But there are many cases where patients have neglected their teeth and had a little pain, or after taking sweets, they have a severe toothache that soon passes off, so that at the time of examination there is no soreness nor inflammation; or if we have a strong, robust, healthy patient, and in excavating carefully around the pulp we remove a layer of decalcified dentine and find just a point of the pulp exposed; in such cases I do not hesitate to cap, and my mixture is as follows:

I take oxide of zinc, and mixing it with oil of clove or creasote, flow it carefully over the pulp, then fill out the cavity with either oxychloride, or oxyphosphate of zinc.

It is true, Mr. President, that I have kept a record of the pulp exposures and my method of treating them, and the condition that they were in, but in looking over my records I found that when I wanted to get at the results, it was not a very easy matter, and it has taught me in the future to keep a little book and enter the cases of pulp exposure and treatment of dead teeth, so I can turn to a person's name and find the result at once, but, you come to look through your record-book for a case which was treated twelve or fifteen years ago and follow out the record of the patient to find whether the pulp has since died, you will have an endless job. I have one case where I had five exposures of pulp in one mouth, of a right superior molar, mesial cavity, and of the bicuspid on the same side, and the bicuspid of the left superior. They were capped eight years ago, and last year—I have not seen the patient this season—they were all tested with warm instruments or ice, and every one of them was alive. How soon they will die, of course, I cannot tell, but they were apparently in good condition when I last saw them. Another case of which I have a record, was an exposure on the mesial surface of an upper right molar, which was exposed in such a manner that with a magnifying-glass you could look into the cavity and see distinctly the pulsations in the pulp. At that time I was with Dr. Shepard, and I called

him to see the case. It was extremely interesting, more so perhaps to us than to the patient. As the patient was strong and healthy, the pulp was capped in the manner I have just described, and that pulp is alive to-day.

That is all the data that I can give you, Mr. President, excepting that I know, in my own practice that a majority of the pulps that I have treated and capped in that way are still alive, and I am a believer in capping certain exposed pulps. This is an exception: A patient came to my office something over two years ago. On a lower right second bicuspid was an exposure near the margin of the gum, which was capped, and that patient had not been in to see me since that day until yesterday, and would not have come then had there not been a pain in that locality. In examination I found that decay had started below the gutta-percha, upon the cervical wall, which I had placed there for protection, after the first capping, and had again exposed the pulp. The patient had suffered for two or three weeks with neuralgia before coming in. I thought it was useless to try to preserve any exposed pulps for that patient, so a new opening was made and the pulp devitalized.

I do not hesitate, however, in destroying a pulp to-day so much as I did some years ago, when we were taught to believe that the devitalizing of an exposed pulp was almost a certain forerunner to the total destruction of the tooth. There has been a radical change in that theory, Dr. Atkinson, of New York, has written a paper in which he speaks of the amputation of a pulp. If I remember rightly, he claims that you may amputate a pulp up to the bifurcation of the roots, that it is good surgery and it will live. I do not agree with him; and yet when I have been trying to destroy pulps up in the canals, I have sometimes thought they would live, no matter what you did to them. I do not hesitate to destroy pulps to-day, because we have such an excellent knowledge of the structure of the tooth, and we now know that simply because the pulp is lost we must not suppose the tooth to be entirely dead, but there is still life from the membranes of the root. So I am now inclined to cap only those pulps which present the best conditions for capping. If a pulp presents any symptoms of congestion or inflammation, I believe you had better dispatch it at once, in order to give the best results to the patient, but I am not a believer in the wholesale destroying of pulps. I believe, gentlemen, in conservative treatment.

DR. EDDY:—Mr. President and Gentlemen: I am a firm believer in treating exposed pulps, with arsenious acid. I have tried almost everything else

during the last fifteen years and have found nothing that will especially preserve the life of a pulp. I have had cases apparently benefited by some kinds of treatment; but I think it has been due fully as much to the temperament of the patient and general condition of health. I believe that oxyphosphate will destroy a pulp just as surely as arsenic, only not as soon. I believe this because I used it nearly eight years, and have seen some of the results. I have also had cases where I have used oxyphosphate of zinc underneath amalgam fillings, as a cement, a non-conductor, but not in direct contact with the pulp, and the pulps have died, and I laid it to the oxyphosphate of zinc. Of course, I have had my experience with fresh exposure, which I suppose was due to my being a graduate, and in those cases I think I have had good results by doing nothing more than touching the exposed pulp with carbolic acid and packing gold directly over that, and it seemed to work very well. I have tried almost everything that has come before us; and in those cases of exposure resulting from decay, or even in cases where there is no exposure, but a zone of softened tissue remains, which is liable to break down, I think it will give the patient the most satisfaction to devitalize the pulp. It may be a little more painful at first, but it is better than having the neuralgia every now and then for from one to five years and then destroy the pulp.

Another thing, Mr. President, when a pulp is destroyed by an oxyphosphate filling it makes one of the dirtiest heaps of debris that I have met. It goes all to pieces and makes a very bad mess. I have used oxychloride for capping and had better results than with the oxyphosphate.

DR. BLAISDELL:—I would like to ask if Dr. Eddy has ever used oxysulphate?

DR. EDDY:—I have not.

DR. GILLETT:—I would like to hear more about this idea of oxyphosphate destroying the pulp. As I understand the gentleman, he makes the statement that he has ascribed the death of certain pulps to oxyphosphate used either as a lining, or as a filling in cases where there has been no exposure. Am I right?

DR. EDDY:—Yes, sir. I had one to-day, Mr. President, that I can attribute to nothing but oxyphosphate. There was no exposure.

DR. NILES:—Mr. President, I used to cap pulps, but I seldom perform that operation now. When I do, I feel that death will occur sooner or later. I never cap with oxyphosphate unless I wish death to take place. It has been said by one of the speakers to-night that the nerve has no means of absorbing dead matter, as the pulp has no absorbents. Even if it had, I should not want it to come in contact with the oxyphosphate. Phosphoric acid is an escharotic, and I don't know why a creamy mixture of it and the oxide should be flown over an exposed pulp, unless to destroy it. If a pulp stands this treatment and lives, it is tough. An acid that will etch glass, or dissolve silver, will destroy soft tissue. It seems to me out of reason to use such things on exposed pulps. Phosphates or chlorides mixed to the consistency of cream make an unstable, strongly acid compound, that when hard very soon dissolves in water, or to a large extent will become deteriorated in water or moisture. In my opinion, the best treatment to save a pulp, provided the patient is strong and robust enough to help out any treatment, is to relieve it from all irritation, disinfect it, and let nature take care of it. Any non-irritating mechanical means you can use to that end will be more beneficial than stimulating, bleeding, etc. I would not use strong carbolic acid or creosote to disinfect an exposed pulp, for they also destroy its surface, and the dead matter remaining decomposes and irritates the adjoining tissue. Sooner or later that pulp dies. The most of the patients who come to us with exposed pulps have a debilitated state of the system. It is not the best vitalized class of people, but those who are sickly and delicate, and who have not much reserve force for the body, to say nothing about the teeth. The question of conservative treatment, in a large majority of cases, with the hope of any lasting results, I shrink from. If I can painlessly treat my patient and take him past the trouble by destroying the pulp and filling the cavity and canal, I feel that I have done the best that can be done for him.

DR. TAFT:—In my own practice I think I have been reasonably fortunate in saving what exposed pulps I have had to deal with. I suppose I have my proportionate share of them, and my practice has been almost invariably to cap them. I think I am safe in saying that I have not used arsenious acid half a dozen times for the purpose of destroying pulps in the four years that I have been out of the school—I do not think I have used it *four* times. In those cases where I have used it, the pulps have come to me in a highly inflamed condition, have been aching a long time, and trying to save them

by capping has seemed at once to be out of the question, and in those cases—and in those cases only—have I devitalized them at once. I do not believe in flowing oxyphosphate directly over the pulp; neither do I believe it a good plan to put carbolic acid over it. My method is either to mix oxide of zinc with oil of cloves and apply it carefully, or to apply the oil of cloves directly to the exposed portion, then dust the oxide of zinc on that; afterwards flowing oxyphosphate in a creamy condition, and waiting until it hardened, then fill the rest of the cavity with oxyphosphate or amalgam.

I keep a record of all the pulps that come to me exposed, or that I expose myself, and some I treat differently from others, but the majority of them, when seen subsequently, are alive. Occasionally I find one that comes back after a time, showing signs that the tooth is dead, and of course the tooth is then opened and treated; but the proportion of them is so very small that I believe in a conservative treatment of them, and think the first thing to be done is to get rid of whatever congestion there may be, if any, and I do not see why a pulp should not then live as well under a careful and skillful capping as it should against the hard, bony wall of the pulp cavity. I have had such good success in the treatment of them that I do not believe in destroying them at once.

DR. GILLETT:—Will Dr. Taft please to outline his treatment of congested pulp, by which he brings it back to normal condition?

DR. TAFT:—I must say that I do not have a great many cases of congested pulp, but where I do have one which has been exposed for some length of time, and has given pain, I first apply local treatment, using something of the nature of oil of cloves, then fill the cavity with cotton and let it go for a few days. I had a case about three weeks ago of a tooth that I had filled, a year ago, with gutta-percha. At the time the tooth was filled I was unable to decide whether there was an exposure or not. The tooth had been previously filled by a well-known Boston dentist with gutta-percha, and was very sensitive to excavating. The cavity extended far up under the gum, and I refilled it with gutta-percha and it kept quiet for about a year. Three weeks ago the lady came to me, complaining of great pain. I took the gutta-percha out, and upon re-excavating I exposed the pulp and found it alive. It had been troubling her then for about a week. I thought at first I would try to save it, and began treatment by putting a pledget of cotton, dipped in oil of cloves, into the cavity, sealing it up temporarily with gutta-percha, and

applying the little capsicum plasters to the gum, hoping to reduce the inflammation in that way, but after working over it a week I destroyed it. I think congestion of the pulp can be reduced by internal treatment much better than by any local treatment, and if we were physicians and had the knowledge of materia medica that we should have under those conditions, I think we would find it possible to easily and quickly get rid of the inflammation and bring the pulp around to a healthy condition, so that it could then be successfully capped, provided there had been no suppuration. When it has reached that stage, the only thing to be done, in my opinion, is to apply arsenious acid and destroy it.

DR. WERNER:—I look upon the treatment of a pulp principally from the standpoint of whether such pulp is absolutely necessary for the welfare of the tooth. The pulp is a formative organ. Its function decreases as the tooth develops. In young persons it is largest; grows smaller and smaller, and in old age it is nearly ossified or obliterated; that is, its retrogressive stage begins materially after the crown of the tooth erupts through the gum. I have seen three different times what Dr. Smith spoke of—the actual pulsation of the pulp—which in a large exposure, in a favorable light, is easily seen. All of those three pulps were capped and they seemingly are alive to-day, but I think the teeth would be quite as serviceable had the pulp been destroyed. From a surgical standpoint the amputation of part of the pulp may at times be successful, for we know how resistive they often are to any kind of arsenious acid treatment, yet I should never do it unless it were in a partially developed tooth, where the life of the pulp is essential. What the surgeon does after trepanning the skull, we should do in capping or bridging over the pulp cavity. He does not flow escharotics like oxychloride or oxyphosphate of zinc over the brain. He mechanically covers and takes good care not to press on the contents of the cavity. The tooth pulp must be treated in the same way; a simple mechanical covering over it being all that is necessary. Whether you put a metallic cap, or whether you put court plaster, or one of gutta-percha (though I should hesitate about using the latter), makes little difference, only be sure that you do not press upon the pulp. With the knowledge we have to-day in treating devitalized teeth, there seems little reason for a young man to risk trying to save bad cases of exposed pulps. It is in extreme practice successful only for the time being—sooner or later ending in failure. To me many so-called successful records are only apparent successes, the operations only

hastening the pulps to degeneration. The flowing over of all escharotics, I think, is highly unscientific.

DR. CLAPP:—The last speaker has made the statement that after the tooth is formed it no longer requires the presence of a pulp. I would like to ask him when he considers a tooth formed?

DR. WERNER:—In many cases the pulps of teeth are very large at twenty-five years. A tooth, however, is usually formed and the apical foramen closed up at from three to five, or, at the latest, ten years after the eruption.

DR. CLAPP:—I had occasion yesterday to cut off two central incisors for a lady eighteen years of age. The pulp chamber was entirely obliterated, entirely filled. I imagine that the pulps near the apex of the root are still alive, and it seems to me that the presence of the pulp is a great advantage to the teeth as long as they are in a healthy condition, and I must say that I hesitate to destroy pulps. As Dr. Taft says he has not destroyed the pulps in half a dozen teeth in four years, I do not believe that my proportion is greater than that in eighteen years.

I would like to inquire of the Society its opinion of the advisability of removing softened dentine over the pulps that would be undoubtedly exposed by the removal of such softened dentine.

DR. BIGELOW:—I do not rise to relate any experiences in capping pulps, but there is one question that has come up in my mind many times that I would like to have answered, if possible, by some of these gentlemen that believe in capping, and practice it regularly. We were taught in the school that in an inflammatory condition we might expect to find heat, pain, redness, swelling—we might expect to find one or all of these symptoms in an inflammatory condition of the tissue. It is said of the bones, that an inflammatory condition might exist and yet have neither swelling, redness, nor pain, but there might be heat. Now the question that comes up in my mind is, how are we to know, really, when the pulp is in an inflamed condition, and what treatment to give it before capping? Very often patients come to us with teeth that have been aching, and there is perhaps a swelling of the pulp and a certain redness; I say swelling—the throbbing generally indicates a swollen condition, in which cases the pulps are evidently inflamed; but what I would like to know is, how are we going to tell whether these pulps are in an inflamed condition or not? I would like to

know the proper treatment for an inflamed pulp. In cases where I have attempted to subdue the inflammation, I have not had great success.

DR. SMITH:—In the absence of all symptoms, I should conclude that there was no inflammation.

DR. BIGELOW:—I don't know whether all the symptoms were absent or not, and that is just the point; there may have been heat, but not knowing positively there was an increase in temperature, an inflammatory condition might exist, and we might work blindly.

DR. SMITH:—I know of a prominent physician who treated a woman for the "Grippe" when her real trouble was fatty degeneration of the liver. He worked as near to the symptoms as he could get, but he diagnosed the case wrong. He is a prominent, eminent physician, and I simply speak of it to show the uncertainty of symptoms in a diagnosis. So in this case, there might be inflammation, but if there was an absence of all the symptoms, I should conclude that there was no inflammation and should treat it accordingly.

DR. BIGELOW:—Suppose there was an absence of all the other symptoms, how are going to get at the heat?

DR. SMITH:—Well, I don't know of any thermometer that has been made to register the amount of heat in a pulp. The only way you can get at it is by the symptoms which the patient gives you. Of course we can suppose lots of things, but we must act on the actual facts presented.

DR. CLAPP:—The gentleman has answered Dr. Bigelow. I would like to have him give his opinion on my question.

DR. SMITH:—Mr. President, it is hardly fair to consider me the encyclopedia of the Society, especially when we have the professor of materia medica of the school here. He ought to be able to answer all these questions.

I don't speak as an authority, but if decalcified dentine is soft and punky, I don't believe in leaving it. I believe you had better take it out and destroy the pulp at once, but if by decalcified dentine you refer to dentine that can be easily removed with an excavator, and yet has a certain amount of hardness about it, I should leave it rather than expose the pulp. I had a case—the patient was a child—where this soft dentine was left in the cavity. It was treated by a dentist whom I knew, and at first I was surprised that he

should have left it, but when I found what a subject he had to deal with, I did not blame him; in fact I think I can compliment him for getting out as much as he did. I took occasion to speak to him about it, and he said he could not do much with the patient, so he left this punky condition, treated it with antiseptics and put in an oxyphosphate filling over the surface; but yet it set up an irritation, and it would have been better if the tooth, a six-year molar, had been extracted at the time. As the better way out of it, I had the tooth extracted, and the twelve-year molar came right into place. So if you mean decalcified dentine of that nature, I believe in removing it and destroying the pulp.

DR. CLAPP:—The gentleman has answered my question completely; I did mean just this leatherly condition that we often find in the deep portions of the cavities, and it is just this soaked condition of the dentine that I wanted his opinion concerning the advisability of removing it, but I referred more particularly to the teeth of adults than of children.

DR. SMITH:—Now that I have given the gentleman my views on the subject, I would like to have his.

DR. CLAPP:—I have left this softened dentine in a great many cases, and so far as my experience goes, I have not afterwards had very much trouble. I think that, rather than expose a pulp in a patient of good health, the surrounding conditions being favorable, I should prefer to leave a small layer, disinfecting and drying as much as possible before filling. It appears to me to be a great calamity whenever I expose the pulp in this way or in any other way, and I do the best I can to avoid doing so. It has been my practice, where I thought I should expose the pulp, to leave this softened dentine, but I have seen a few cases where decay has undoubtedly continued underneath the oxyphosphate or oxychloride fillings. I think it is a matter of great importance, and one where the best judgment will sometimes go astray.

DR. EDDY:—I would just like to mention that hydronaphthol may be fearlessly used in all cavities without causing any escharotic action.

With regard to the spongy condition that Dr. Clapp speaks of, it must be thoroughly dried out with a hot air syringe. If it cannot be thoroughly dried, it is better to remove it.

DR. WERNER:—In regard to the question asked by Dr. Clapp, I should say that it would depend on the amount of sensibility yet left in the partly decalcified leathery part of the dentine spoken of. If that still has sensibility, I should believe in treating it not so much escharotically as antiseptically. Again I wish to assert my opinion, that after the foramen was closed and the tooth well developed, it would not be a great calamity for a tooth if the pulp were devitalized and the pulp cavity well filled. I am much of the opinion that a tooth in a devitalized condition resists better than a live tooth the attacks of caries. This is perhaps an extraordinary statement to make, and I know is contrary to the teachings of good authority, yet the many pulpless filled teeth I see free from decay, while their living neighbors do decay, leads me to this opinion.

DR. REILLY:—I would like to state a little experience that might cover Dr. Clapp's case partly. Some time ago a young lad came to my chair whose father had great difficulty in getting to go to a dentist's office. There were two enormous labial cavities in the central incisors. I did not attempt to excavate at all. I had been cautioned before that it would be hazardous to try it, as he was a fearful little fellow to get along with, and his teeth were very sensitive. I simply wiped out the cavities as well as I could, getting them dry without causing pain, and filled them with gutta-percha. He came again a few weeks ago, and the dentine was all there, much harder, as I remember it, than it was on the first visit. I removed it the last time and filled them without any trouble. Now, I do not think it is any risk to leave decalcified dentine in the bottom of a cavity, providing it is tightly sealed; but, as Dr. Eddy says, I should depend more on the drying process than on the use of antiseptics.

DR. WERNER:—Dr. Reilly brought up a very important point. Much less excavation than is generally done is necessary for the absolute control of decay, and that all excavation that gives pain is, in one sense, unscientific. It is only necessary, from a mechanical standpoint, to retain metallic fillings, but I think some day we will have a plastic filling that will require very little excavation—simply the wiping out of the cavity and the thorough removal of the actual decayed portion, leaving all that has sensation—treating the cavity antiseptically, and when in an aseptic condition, fill it with this plastic filling. The surgery of to-day is very much modified. Muscles and bone are not cut away as they were forty, or even twenty years

ago. To-day ivory is inserted and made useful in the place of bone, and why should not tooth substance that has sensation be preserved? We have to cut it away simply because we have to shape the cavities to retain the metallic fillings; that is, we treat mechanically instead of therapeutically. In the near future I hope we shall know better than to cut away that which should be saved. I have a very hard little patient whose parents wish me to save his teeth but not to give him pain, and the little fellow will not stand any excavation. His teeth are being preserved at the only disadvantage of having to have a great many temporary fillings.

DR. SMITH:—I do not quite understand the logic of Dr. Reilly's conclusions. I may misunderstand him. He tells of a case coming to him where he did not see fit to excavate; he placed in that cavity some gutta-percha; the gutta-percha after being in a while was removed and the cavity excavated. Now, how did he know the depth of decay at the first sitting? and not knowing the depth of the cavity, how did he know that decay had not gone on in that cavity?

DR. REILLY:—I think I was cautious about that statement. I said, to all appearances, it had not decayed. Of course, I could not tell absolutely, only from my experience in excavating I judged it was about the same as when I first saw it, perhaps a little harder. If we were not able to judge something of the depth of decayed dentine, we would be in constant danger of exposing the pulp. If you cannot tell by the feeling of your excavator when you are near the pulp, I would like to know what you are going to do in those cases where you cannot see the bottom of the cavity?

DR. SMITH:—Yes; but you say you did not use your excavator in this case.

DR. REILLY:—I reached my conclusions while wiping out the cavity with the forceps and cotton, and I thought then, and it is my feelings now, that decay had not gone on in that cavity at all. I think it is possible for such a condition of affairs to exist; I think that things stand still in such a condition. I had a case the day before yesterday which was treated some ten years ago, and on removing the filling I found the pulp canals filled with cotton. There was no odor at all—perhaps the cotton was slightly discolored, but the cavity was in apparently the same condition in which it had been left; and it seems to me that if a tooth would remain in that condition with cotton, why wouldn't it be preserved under gutta-percha in

just the same condition? I don't know why the gentleman should conclude that it would decay? What leads him to think that it would decay?

DR. SMITH:—I don't know that it is so, but I have read somewhere that in decayed dentine there exists "pesky little bugs," and if a cavity is stopped up and these micrococci, or bacteria, cannot get at the oxygen which comes to a cavity from the outside, they will work in to the dentine and decalcify the tooth substance in the search for that food which they live on. It is not my theory, but I read it somewhere, and in my practice I either kill the bugs or take them out.

DR. REILLY:—That theory may be all right, but it does not work out in my practice. The experience that I had with this young man has been repeated over and over. I do not excavate carious dentine as far as it can be, and I don't think that theory will work out in all cases. If bacteria are there after the cavity has been thoroughly cleansed and tightly stopped, they must find an early death in some way.

DR. NILES:—I am very glad to hear Dr. Reilly speak as he does. It is perfectly consistent with the accepted theory of decay. Decay does not progress where moisture and oxygen are excluded; therefore a cavity thoroughly dried out and hermetically sealed is good treatment and sound in theory. Nature has formed a covering to the nerve, and although nothing but the matrix of that covering may remain, I would not tear it away if an attempt is to be made to save the nerve. I would retain it with its odontoblastic connections with the pulp. I believe it is the only reasonable theoretical and practical course to follow. Let the nerve alone; nature has provided a covering, why should it be torn away? If there is any irritation, it is very easily removed. In my opinion, a pulp is never inflamed to an extent that needs treatment when it is free from soreness by percussion. If there is inflammation at all in the pulp there will be periosteal disturbance to give warning; but if the periodontal membrane is not inflamed, it may be concluded that the pulp is free from irritation; provided, of course, it has never been painful or previously inflamed.

DR. REILLY:—I would like to relate my experience with cases of this kind which have been right under my eye and care for some two and one-half years. When I first began to fill my babies' teeth, I made it a point that they were not going to be frightened. The eldest is six years old, and has been

under treatment for nearly three years, and during that time she has never experienced any pain in the chair, and she will come to my office any time with confidence,—in fact she seems to like it. Her teeth are very poor, and in the first ten cavities I did not do any excavating whatever. I practically did not remove any decayed dentine. I simply stuck in fillings wherever there was a cavity and depended entirely on dryness, and I have had good results from it, with no dead pulp up to the present time. After the first dressing of cement I endeavor to remove carious dentine.

DR. TAFT:—Did you use a rubber dam?

DR. REILLY:—No, sir. The teeth were too short and it could not be done without causing some pain. My principal object was to avoid pain.

DR. WERNER:—Did you put in any gutta-percha fillings?

DR. REILLY:—Yes, sir. I put in some at first; then cement, and I have used copper amalgam in crown cavities.

DR. GILLETT:—It seems to me that the case which Dr. Reilly has related is merely the ordinary temporizing with children's teeth, the object being to save them pain, and the case would be somewhat different with adult teeth in which there has been a large amount of decay, and when we expect to do permanent work. I do not see how that decayed, diseased matrix can be non-irritating, as Dr. Niles claims.

DR. NILES:—If it was irritating you would have inflammation.

DR. GILLETT:—A pulp will sometimes stand much irritation without showing evidence of inflammation. An inflamed pulp is not always painful. My experience with decalcified dentine under fillings has been gained chiefly from removing fillings which other dentists have inserted over such decalcified dentine, and I have seen many cases where the results were anything but satisfactory.

DR. NILES:—I have had some experience in that line myself during the last thirty years, and I have generally found decalcified dentine under good fillings to be in as good condition as when it was left there. In most cases it was left there, not with any idea of conservative treatment, but because the dentist was too lazy to take it out. I do not believe that decalcified dentine, thoroughly cleansed and dried, and hermetically sealed, can do any harm whatever to a nerve, or create any irritating effect. I would not leave a

sloppy piece of dentine in the bottom of a cavity, which could not be well dried; neither would I leave the cavity half-full of decay, because in that case I could not reasonably expect that there would be no irritation.

DR. EDDY:—If my memory serves me right, Dr. Niles once read a paper before the American Academy of Dental Science, in which he stated that in those cases of decalcified dentine in the bottom of the cavity he invariably used arsenious acid, and treated them the same as he would an exposed pulp, and it went on record. Now, to-night, he seems to be advocating an entirely different method.

DR. NILES:—I should like to see that statement. I have written a paper on the subject—you will find it probably in the *Independent Practitioner*—but I don't think you will find the statement there that I always destroy the nerve in cases where there is decalcified dentine at the bottom of the cavity. Of course, where I have inflammation in the pulp and the tooth is sore, I should destroy the pulp.

DR. BRIGGS:—Rules have their exceptions, and when I make the statement that I do not believe in capping pulps, I still claim the privilege of having certain exceptions. It seems to me that in all the cases of actual exposure, the question of capping the pulp is merely one of expediency for the purpose of keeping the tooth along. You do not wish to destroy the pulp at that time, perhaps, because it is a young tooth; or perhaps you dread the trouble and inflammation which is likely to ensue from the use of arsenious acid at that time, or possibly the patient is going away. In those cases I have had no trouble in putting a dressing in, which has kept the tooth quiet for a long time, but I always expect that sooner or later I will have a dead pulp to attend to. In fact, the most of the pulps that I have capped to preserve, are pulps that are not exposed, but are protected by a thin layer of dentine, perhaps decalcified, but which I could easily make aseptic. Many cases of pulp stones are the result of putting a filling too near the pulp, which is irritating and causes a formation. In those cases I remove the irritant and put in a dressing. That dressing is made by mixing the oxide of zinc with an antiseptic which has some anæsthetic property—say some essential oil; then covering that with a thin layer of copper plate, and then putting a hard filling over that. There is always an uncertainty about it, and four, eight or even twelve years is not always time enough to prove the success of capping the pulp. It may be that after twelve years you find a pulp stone

formed that will give all the trouble and exhibit all the symptoms of neuralgia. If your tooth is fully formed, it is better to get the pulp out of the way and to fill it properly, and have a good, healthy tooth. It has been said this evening that the office of the pulp is to form the tooth, and when the tooth has been fully formed its usefulness is, to a greater or less extent, over, and the tooth can do good service without it, and this opinion I agree with.

Perhaps right here would be a good place to mention something which I intended to speak of later on, under the head of "Incidents of Office Practice." We have, all of us, more or less trouble from applying arsenious acid, and I have lately found that where the pulp is exposed, I can, by wiping a 20 per cent. solution of cocaine over the surface, inject without pain a twenty per cent. solution of cocaine, and after a few moments remove the pulp entirely, also without pain. My brother and myself have operated several times on such cases lately with satisfaction to ourselves and our patients. I say several cases, because we have disposed of some old cases which had resisted the arsenious acid and had been hanging along waiting for "something to turn up."

DR. GILLETT:—Dr. Briggs' remarks concerning injecting cocaine solution, and removing pulps in this way, brought to my mind that the chief use that I make of cocaine is in connection with arsenious acid, when I am going to destroy a pulp. My method is to moisten the end of an instrument, and pick up with it some of the cocaine crystals, making a pellet the size of two or three pinheads, and seal it in with the arsenious acid. I find it very helpful in controlling the pain in connection with such application.

PRESENTATION OF SPECIMENS.

PRESIDENT COOKE:—Dr. Gillett will show a warm-air apparatus.

DR. GILLETT:—I was prompted by the note on the card, saying that an apparatus for obtunding dentine would be shown, to bring up this apparatus, which some of you have seen before, as Dr. Brackett exhibited it at the last meeting of the American Academy of Dental Science. This is an apparatus for obtunding sensitive dentine, of which there are but a few of its kind in the country, and this particular one was sent to Dr. Brackett by Dr. Bogue, of New York. I have used it some for the last two or three weeks, and find that it is possible to produce very good results without hurting the patient.

Those who have used Dr. Waite's obtundent, and have gotten satisfactory results with it, will perhaps appreciate my explanation in this way. I have used that with considerable success, and like it very well, and I find that I can do the same work with this apparatus, with the additional gain that its application is not painful, as a rule. It is simply a means of obtaining dryness in the cavity by the use of a continuous current of warm air.

H. L. UPHAM, D.M.D.,

Editor Harvard Odontological Society.

PHOSPHATE OF ZINC CEMENT AS ANCHORAGE FOR PERMANENT FILLINGS.[\[2\]](#)

BY C. J. PETERS, D.D.S., SYRACUSE, N. Y.

About six years ago I read a paper before the Syracuse Dental Society on the subject, "Oxy. Phosphate," in which my main object was to bring forward the idea of its use as anchorage for amalgam fillings in particular. After the experience I have had with the method, I have nothing to retract, but can reiterate with emphasis every word said then. This, now, is no new untried thing I bring before you. For the last two years it has been talked of more or less throughout the country. I conceived the idea early in 1884 when a lady came to me for treatment in whose mouth were four teeth which two years previously she had been told were beyond saving, but had neglected having them extracted on account of the dread of the operation. I treated three of these teeth and filled the roots, but could think of nothing that would be retained in the crown but cement. I wondered if amalgam would stick to cement as the cement does to tooth structure. I tried it, and was successful, and those three fillings are in to-day and can be produced at any time.

The good results attending its use with amalgam, suggested at once numerous uses to which it might be put, as: anchorage for gold, foundation for any other filling at one operation, lining for thin walls when objectionable color of filling material could show through, cementing fillings which had fallen out or loosened intact, last to place temporarily or

otherwise. As an anchorage, its greatest is, I believe, under amalgam fillings in cases of badly broken down molar and bicuspid crowns where, on account of excessive decay and nearness of the pulp, reliable undercut cannot be obtained, and if it could be, would so weaken what remained of the crown as to deprive it of strength to carry the filling. It is especially useful in molars where the buccal surface and a half or more of the grinding surface is gone, in bicuspids whose proximate surfaces are gone with a large share of the grinding, so that the opening of the cavity is the full size of the circumference of the tooth and the remaining walls thin. It is astonishing how many of these latter cases can be shaped and contoured with amalgam without a particle of the amalgam being in sight after the work is finished. If in an anterior proximal filling proper contouring make it necessary for some of the amalgam to show, the anchorage may be thoroughly relied on to permit of cutting out a small portion and facing with gold. The shading of the thin walls by the amalgam is absolutely prevented by the cement. I have in my own mouth a molar in which the filling was loosened by mastication five times on account of the cavity being one of the kind where reliable anchorage could not be obtained. The pulp finally became exposed from frequent cutting away of tooth structure, and the sixth filling was anchored by taking advantage of the pulp chamber, or a portion of it. To use cement as anchorage for amalgam, this is the process: Mix the amalgam according to your custom; place on a slab for mixing cement a small quantity of the liquid and powder ready to mix; then, the cavity being prepared, dry it thoroughly and keep it so while mixing the cement. This being done, place a small amount in the cavity, and at once upon it place a piece of amalgam, which should be so manipulated with the instrument suited to the size and shape of cavity as to force the cement under the amalgam all over the floor of the cavity. Care should be taken to not force the cement entirely to the cervical edge in proximal cavities, and any excess of cement used should be worked out at a point easy of access. By this time the cement is hard enough to be easily chipped off whenever it has been forced beyond the inner edge of the enamel. The filling with amalgam is proceeded with and finished in the usual manner. It is essential that the cement should not be smeared over the edges of the cavity, but carefully worked all over the dentine, closing the mouths of the tubuli, allowing the enamel edge free for contact with the amalgam. The edges of such a filling are better, and remain better than those of the ordinary filling, for the

reason, I believe, that the cement controls, or at any rate lessens, the spheroidal tendency in the amalgam. In very deep cavities it is well to work into the cement a piece of hardened amalgam before inserting the fresh, as it lessens the amount of cement necessary, and also of the amalgam, and again prevents the tendency to spheroid. Proximal cavities in children's teeth are very easily filled with amalgam by this method, and without causing pain in cutting tooth structure.

We now come to the use of cements as anchorage for gold. With this material it has a smaller field of application, and in my hands does not yield as good results as in the use of amalgam. I believe that Dr. F. D. Nellis, of Syracuse, New York, was the first to conceive the idea and use the cement as an anchorage for gold. It is useful in teeth having very shallow cavities, and in those cases where, while it is desirable to use gold, the edges chip or shale off at every attempt to make the cavity retentive. The method of use is as follows: The rubber dam, of course, is used. The cavity being ready, a small amount of cement is mixed and placed in it. On the cement put a cylinder of gold large enough to cover the floor of the cavity. Work the gold into the cement, at the same time working the latter all over the cavity. Trim cement from edges and proceed with the filling, making a mechanical anchorage of the gold with that anchored by the cement. Foil or pellet may be used in place of cylinder, and it may be cohesive or not, and the filling finished with soft or cohesive gold, but I think the best results are obtained by using soft cylinder over the cement, continuing with the same and finishing with a few strips of annealed foil. A very good way is to proceed to fill the cavity with gold the same as if cement were not to be used for anchorage, holding the gold in place with another instrument, and when sufficient has been inserted to nicely take the form of the cavity, to take out, place a little cement in the cavity, then force the gold back to place, and after waiting a minute or two for the setting, go on and finish the filling. It is not necessary in this work to confine oneself to the use of the phosphate of zinc cement. In restoring color to very dark pulpless teeth, and wherever extra whiteness is desired, the oxychloride may be used with advantage, but must not be relied upon for strength like the oxyphosphate. In shading, I get the best general results from the yellow shade of the latter. Pulpless teeth generally have a bluish tinge, and yellow seems to neutralize it very effectually. In regard to the effect of oxyphosphate in deep cavities in teeth with living pulps, it does not seem necessary for me to say much here, since

the subject has long been worn threadbare by the profession. However, it must be borne in mind that here is a difference. The cement in use I have described is sealed away from the fluids of the mouth and does not have the same effect as when being disintegrated by those fluids. It remains a perfectly inert substance so long as moisture is kept from it, and has the same use in the floor of the cavity as the varnish so often recommended for closing the tubuli of the dentine to prevent ingress of moisture from that direction. In such cases it is a very simple matter to touch the bottom of the cavity with liquid gutta percha before using the cement. In cases of exposure where capping is desired, cap according to your custom and then proceed as described. When extirpation is resorted to, I fill root and pulp chamber with gutta percha.

CAPPING EXPOSED PULPS. [\[3\]](#)

BY A. H. FULLER, M.D., D.D.S., ST. LOUIS, MO.

I will, in a brief manner, give something of a history of the practice of capping the exposed pulps of teeth, with a view of rendering the teeth comfortable and useful, and at the same time preserving the vitality of the capped pulp.

In the early days of the profession, up to and including the time of Hunter, Fox and Bell, the exposure of the pulp was almost equivalent to the absolute loss of the tooth; not from the fact that dentists were ignorant of the conditions necessary for its retention, but from the want of instruments and appliances with which to perform operations for its preservation.

We find them attempting in various ways to overcome the difficulties with which they were surrounded. They attempted to shrink the pulp and render it less sensitive, that they might fill over without wounding or pressing upon it. To accomplish this, astringents and opiates were resorted to; again, acids, alkalies, or the actual cautery were used to destroy it. Any of the above operations were rarely possible, and when possible, still more rarely successful. The extraction and replantation of the aching tooth was resorted to, the pulp removed and the root canals filled in cases where the means at

hand would admit. Cleansing the tooth by boiling before replanting, was *suggested* by Hunter.

Following the above methods, the operation of excision had its day, its advocates and its opponents. This consisted in removing the crown and as much of the root and pulp as possible with cutting forceps, adapted to the different teeth.

The following from "Waite" would lead us to infer that there were those in his day, as well as at present, who would mount a hobby and attempt to ride into prominence by so doing. In discussing this operation of excision, he says: "At the present period, nothing is aimed at but novelty; nor do many productions succeed that follow the regular line of going on; and while men of science pursue honorable and just means to bring themselves into notice, their reputation is frequently surpassed by cotemporaries, brought forward by some lucky coincidence."

This digression may possibly be excused, as it was, in a measure, necessary, in order to continue the history of attempts at conservation of the dental pulp. I will copy from a few of the earlier writers on dental subjects, to show methods of practice, together with results as given:

Jourdan, writing in 1784, says: "I saw a right canine tooth, in a young lady of twenty-three, so worn away as to expose the commencement of the pulp cavity. I enlarged the opening, gave vent to a drop or two of dark, offensive blood, and the pains, which had been very acute, ceased. I destroyed the sensibility of this tooth with cotton dipped with ether, and filled it with gold. It is still in the mouth and gives no trouble, though it has a blueish hue."

"Koecker" gives the method pursued by him from about 1817. He first produced a black scar by burning, with a red-hot wire, the exposed pulp: covered this with a capping of lead foil, and filling over this with gold. He had previously used gold, and afterwards tin foil, as a capping, but with indifferent success. He attributed the later successes to the cooling properties of the lead.

He relates case after case, and adds: "Five out of six teeth can be preserved alive when this operation is *skillfully* performed." He relates one case where he treated and capped seven teeth, and successfully preserved six of the

pulps alive for many years. He further says the smallest error will inevitably cause the destruction and loss of the tooth. He devotes ten pages of his "Principles of Dental Surgery" to this subject.

"Fitch," 1835, recommends the treatment of the exposed pulp with Aleppo galls, scraped up, placed on the exposure and covered with wax; a few weeks or months later, caps with sheet lead or piece of gold plate and fills with gold. He relates numerous cases and complete success of this practice.

Harris, in 1840, arched his gold fillings over the exposure and continued the filling with gold, with some success. Had tried Fitch's method with indifferent success, and thought Koecker's must from necessity result in failure.

In 1841, "Lefoulon," in his work, after giving some of the methods and remedies of his predecessors and contemporaries, says: "As for ourselves, though we be accused of being controlled by one permanent notion, truth compels us to say that the employment of our ethereal alluminous paste has enabled us to preserve the teeth of our patients, even when these organs were attacked with most intense inflammations. Its sedative and extraordinary anti-spasmodic quality does not fail to triumph over the inflammatory erythema of the dental nervous system and its appendages to such an extent that all pain and all irritation cease at the end of some days, sometimes on the next day. Let the incredulous put us to the proof and test for themselves the truth of our words."

The above quotations will show about the status of the professional opinion as expounded by the authors of the then works on dentistry. Most of these were written by practitioners who claimed special skill, and were in a measure separated from each other and the profession.

With the exception of Hudson, Maynard and, possibly, one or two others, the practice of removing the pulps and filling the canals was not known or attempted. Harris, in 1840, speaks of their operations, but had attempted it only in a few cases, and with indifferent success.

The necessity for, at least, claiming to succeed in operations of this kind—saving pulps—was almost imperative.

In 1850, Dr. J. D. White, in the *News Letter*, says: "The treatment of the exposed pulp has given rise to great difference of sentiment among well

educated dentists, but mainly about the means which should be employed for that purpose, agreeing pretty generally that it is bad practice to destroy it entirely. But as well might we expect to procure a healthy function of the reto-mucosum when denuded of the epidermis by substituting one of our own invention, as to procure a healthy function of the pulp when deprived of its natural protection, the bone."

From about this time until within a comparative recent date, in all methods of capping the ultimate design has been to secure a production of secondary dentine at the exposed point of the pulp. The methods and materials made use of are almost numberless, and the successes claimed are also equally as numerous. For applications in the treatment we have recommended to us in our text-books, pure nitric acid, pure carbolic acid, pure creosote, iodoform, dilute chloride of zinc, iodine, bichloride of mercury, and in fact everything from stimulating it with the electric cautery, to feeding it upon lactophosphate of lime and powdered dentine. For capping: the different cements, stoppings, gums, minerals, etc.; and the cases run from the capping of the remaining third of the pulp in the root of a lower incisor upon which a pivot tooth was to be placed, an exposure in the distal surface of a third upper molar, and this suppurating, to a simple exposure from excavating a cavity for filling.

Our dental societies in their reports show that some are very successful in their endeavors to save exposed pulps: that others don't kill babies for the sake of having a funeral, and attempt to save everything: while others, whose experiences are just as extensive, whose observations and opinions are just as much to be respected, claim that an exposed pulp that has ached or been inflamed, can never become healthy by any treatment whatsoever.

There has been, without doubt, a gradual change in practice for the past fifteen years, brought about by sad experiences and the better knowledge of the tissues involved; also by the improved educational condition of the profession as a whole. Dr. Black, in the *American System of Dentistry*, on "The Pathology of the Dental Pulp," has given us, as far as I am able to judge, by far the best and most instructive contribution upon this subject. He concludes his sixty pages of valuable observations as follows:

"PATHOLOGY OF THE DENTAL PULP—GENERAL CONSIDERATIONS."

"In the foregoing pages I have frequently alluded to the fact—which is apparent in a very large proportion of my microscopic preparations—that any of the secondary calcific formations within the pulp of the tooth, result in exhaustion and final death of the pulp. This fact is so prominent that it seems to me that it cannot well be overlooked; and yet, in the capping of exposed pulps, it seems to have been the thought of the profession that to be able to obtain a secondary deposit under such circumstances was to insure the permanence of the health of the pulp. This was my own thought some years ago, but further clinical experience, combined with closer microscopical study of the subject has convinced me that this is a mistake. Secondary deposits may, and do, insure temporary quiet, but so far from insuring health are they, that, as a matter of fact, they bring about the very conditions that we most wish to avoid—the degeneration and final destruction of the pulp.

"In a large majority of cases, however, this result is brought about very slowly, and thus has escaped the notice of most observers; for if an exposed pulp is capped and the cavity filled, and the case seems to do well for a year or two, it is regarded as a success, and is lost sight of. When this returns some years later with a dead pulp, it is treated as one of the great mass of such cases that are constantly presenting themselves, and probably no note was made of the fact that it was capped at a certain time, and was one of the many successful cases.

"Very many cases of capping pass on for years without any deposit whatever, and seem to remain in a perfectly healthy condition. This we must regard as the most desirable result that can be obtained. Enough of these cases have been noted to demonstrate the possibility of rendering the conditions so nearly normal that no disturbance of the functions of the organ occurs."

From my own experience and observation, and with all due regard for the experiences and opinion of others, I will say, when there has been an exposure of the pulp, with any considerable pain as a consequence, I would in all cases destroy and remove. In cases where from the location of the cavity, or where the surroundings would not admit of a satisfactory operation otherwise, I would devitalize. With the remedies at our command, the increased facilities for destroying and removing the pulp, cleansing the canals, and the various means for hermetically filling them, thereby

rendering the tooth absolutely free from painful impressions, just as serviceable, just as beautiful, and in all respects as enduring. I fail to see how we can justify ourselves, or do justice to the patient by subjecting them to a painful and, possibly, an imperfect operation, with the probabilities that they will again be obliged to seek for relief from the same offending member.

In conclusion, I would add, that accepting the reasoning of Dr. Black as the correct explanation of the conditions presented, I believe there are, comparatively, a small number of cases where we are able to so protect the exposed pulp as to leave it in a normal condition; and while we should endeavor to save the patient time and expense, we should also endeavor to secure them relief from pain and future suffering, and also protect ourselves and the profession from the criticisms following operations that are worse than failures. Since my experience has enabled me to distinguish, to a certain extent, between facts and fables, between the teachings of reasoning intelligence and egotistical statements, my practice has been most satisfactory, my ways have been those of pleasantness, and my paths those of comparative peace.



DRS. J. A. PRICE, Weston, Mo., J. C. Goodrich, Wentzville, Mo., J. H. Kennerly, Lebanon, Ill., P. H. Helmuth, Highland, Ill., Geo. Cameron, Carrollton, Ill., F. A. Green, New Albany, Ind., F. H. Caughell, Morrison, Mo., J. O. Eppright, Odessa, Mo., R. R. Vaughan, Fulton, Mo., visited St. Louis during the past month.



OBITUARY.

At the annual meeting of the Southern Illinois Dental Society, held in Chester, October 21st, 1890, the following resolutions, expressive of the sense of the Society, relative to the death of Dr. Homer Judd, and Dr. M. D. LaCroix were adopted.

WHEREAS, The Southern Illinois Dental Society having learned with profound regret of the death of Dr. Homer Judd since the last annual meeting, the members in convention assembled desire at this time, both individually and collectively, to testify to their high esteem and great love for the life and character of the deceased. Much of the success of the Society is due to his kind helping hand as one of the founders, and the admirable rules and by-laws governing it, were largely the work of his experience.

The members of the Southern Illinois Dental Society will ever hold in reverent regard the memory of Dr. Homer Judd, and extend to his family their respectful sympathy; and the Secretary is hereby requested to advise the family of this action.

C. B. ROHLAND, }
T. W. PRICHITT, }
}Committee.
L. BETTS, }

WHEREAS, We have learned with deep regret and sorrow of the death of our friend and professional brother, Dr. M. D. LaCroix of Lebanon, Illinois, in the prime of his young manhood and usefulness in the dental profession; it is hereby

Resolved, That in the death of this estimable young man, the Southern Illinois Dental Society has lost one of its most earnest and enthusiastic members, the world a useful and honorable citizen, and the social circle a faithful and beloved companion;

Resolved, That our heartfelt sympathies are hereby tendered to the family of our deceased brother, in the hour of their sad bereavement;

Resolved, That these resolutions be entered upon the records of this Society, that copies be sent to the family of the deceased, and to the dental journals for publication.

J. J. JENNELLE, }
C. C. CORBETT, }
Committee.
R. H. CANINE. }

NORTHERN OHIO DENTAL ASSOCIATION.

The thirty-second annual meeting will be held in Oberlin, Ohio, Tuesday, May 12th, 1891, at ten o'clock A.M., and continue its sessions three days.

SUBJECTS FOR DISCUSSION.—"Development of the Teeth." Paper by Dr. W. H. Whitslar, Youngstown. Discussion opened by Dr. A. J. Dowds, Canton.

"The Recurrence of Decay in Teeth." Paper by Dr. J. G. Templeton, Pittsburgh, Pa., and "Hind Sight." Paper by Dr. W. H. Atkinson, New York. Discussion opened by Dr. C. R. Butler, Cleveland, and Dr. E. J. Waye, Sandusky.

"The Sanitary Condition of the Mouth, and How Best to Maintain It." Paper by Dr. J. F. Dougherty, Canton. Discussion opened by Dr. W. T. Jackman, Cleveland and Dr. J. H. Wible, Canton.

Correspondence.

Correspondence.

DEPOSIT PLATES.

EDITOR ARCHIVES:—Please allow me a word in regard to the electric deposit plate. Mr. E. E. Clark, the owner and manager, has been absent in the West in its interest most of the year, and I have, in his absence, undertaken to look after the making of the plates. For a considerable time past, the plates have been sadly deficient in gold, sometimes not enough being put on to properly vulcanize over.

I was unable to account for it, knowing that full quantity of gold was supplied. The secret has been discovered, and the thief lined his pockets instead of coating the plates with gold, and is now in jail awaiting trial for larceny.

The plates as now made are all right in every respect, and it is hoped that all its former friends will again come back to its use.

Yours, etc.,
C. S. STOCKTON.

Newark, Nov. 17th, '90.

Selection.

Selection.

PEROXIDE OF HYDROGEN AND OZONE.
THEIR ANTISEPTIC PROPERTIES. [\[4\]](#)

BY DR. PAUL GIBIER,

Director of the Pasteur Institute of New York.

GENTLEMEN:—Since the discovery of the peroxide of hydrogen by Thenard, in 1818, the therapeutical applications of this oxygenated compound seem to have been neglected both by the medical and surgical professions; and it is only in the last twenty years that a few bacteriologists have demonstrated the germicidal potency of this chemical.

Among the most elaborate reports on the use of this compound may be mentioned those of Paul Bert and Regnard, Baldy, Péan and Larrivé.

Dr. Miguel places peroxide of hydrogen at the head of a long list of antiseptics, and close to the silver salts.

Dr. Bouchut has demonstrated the antiseptic action of peroxide of hydrogen, when applied to diphtheritic exudations.

Prof. Nocard, of Alfort, attenuates the virulence of the symptomatic microbe of carbuncle before he destroys it, by using the same antiseptic.

Dr. E. R. Squibb, [\[5\]](#) of Brooklyn, has also reported the satisfactory results which he obtained with peroxide of hydrogen in the treatment of infectious diseases.

Although the above-mentioned scientists have demonstrated by their experiments that peroxide of hydrogen is one of the most powerful destroyers of pathogenic microbes, its use in therapeutics has not been as extensive as it deserves to be.

In my opinion, the reason for its not being in universal use is the difficulty of procuring it free from hurtful impurities. Another objection is the unstableness of the compound, which gives off nascent oxygen when brought in contact with organic substances. [\[6\]](#)

Besides the foregoing objections the surgical instruments decompose the peroxide, hence, if an operation is to be performed, the surgeon uses some other antiseptic during the procedure, and is apt to continue the application of the same antiseptic in the subsequent dressings.

Nevertheless, the satisfactory results which I have obtained at the Pasteur Institute of New York with peroxide of hydrogen, in the treatment of wounds resulting from deep bites, and those which I have observed at the French clinic of New York, in the treatment of phagedenic chancres, varicose ulcers, parasitic diseases of the skin, and also in the treatment of other affections caused by germs, justify me in adding my statement as to the value of the drug.

But, it is not from a clinical standpoint that I now direct attention to the antiseptic value of peroxide of hydrogen. What I now wish is merely to give a full report of the experiments which I have made on the effects of peroxide of hydrogen upon cultures of the following species of pathogenic microbes: *Bacillus anthracis*, *Bacillus pyocyaneus*, the bacilli of typhoid

fever, of Asiatic cholera, and of yellow fever, streptococcus pyogenes, micro-bacillus prodigiosus, bacillus megaterium, and the bacillus of osteomyelites.

The peroxide of hydrogen which I used was a 3.2% solution, yielding fifteen times its volume of oxygen; but this strength reduced to about 1.5%, corresponding to about eight volumes of oxygen, by adding the fresh culture containing the microbe upon which I was experimenting. I have also experimented upon old cultures loaded with a large number of the spores of the bacillus anthracis. In all cases my experiments were made with a few cubic centimetres of the culture in sterilized test-tubes, in order to obtain accurate results.

The destructive action of peroxide of hydrogen, even diluted in the above proportions, is almost instantaneous. After a contact of a few minutes, I have tried to cultivate the microbes which were submitted to the peroxide, but unsuccessfully, owing to the fact that the germs had been completely destroyed.

My next experiments were made on the hydrophobic virus in the following manner:

I mixed with sterilized water a small quantity of the medulla that had been taken from a rabbit that had died of hydrophobia, and to this mixture added a small quantity of peroxide of hydrogen. Abundant effervescence took place, and, as soon as it ceased, having previously trephined a rabbit, I injected a large dose of the mixture under the dura mater. Slight effervescence immediately took place, and lasted a few moments, but the animal was not more disturbed than when an injection of the ordinary virus is given. This rabbit is still alive, two months after the inoculation.

A second rabbit was inoculated with the same hydrophobic virus, which had not been submitted to the action of the peroxide, and this animal died at the expiration of the eleventh day, with the symptoms of hydrophobia.

I am now experimenting in the same manner upon the bacillus tuberculosis, and if I am not deceived in my expectation, I will be able to impart to the profession some interesting results.

It is worthy of notice that water charged, under pressure, with fifteen times its volume of pure oxygen has not the antiseptic properties of peroxide of

hydrogen. This is due to the fact that when the peroxide is decomposed nascent oxygen separates in that most active and potent of its conditions next to the condition, or allotropic form, known as "ozone." Therefore, it is not illogical to conclude that ozone is the active element of peroxide of hydrogen.

Although peroxide of hydrogen decomposes rapidly in the presence of organic substances, I have observed that its decomposition is checked to some extent by the addition of a sufficient quantity of glycerin; such a mixture, however, cannot be kept for a long time, owing to the slow but constant formation of secondary products, having irritating properties.

Before concluding, I wish to call attention to a new oxygenated compound, or rather ozonized compound, which has been recently discovered, and called "glycozone," by Mr. Marchand.

This glycozone results from the reaction which takes place when glycerin is exposed to the action of ozone, under pressure—one volume of glycerin with fifteen volumes of ozone produces glycozone.

By submitting the bacillus anthracis, pyocyaneus, prodigiosus, and megaterium to the action of glycozone, they were almost immediately destroyed.

I have observed that the action of glycozone upon the typhoid fever bacillus, and some other germs, is much slower than the influence of peroxide of hydrogen.

In dressing of wounds, ulcers, etc., the antiseptic influence of glycozone is rather slow if compared with that of peroxide of hydrogen, with which it may, however, be mixed at the time of using.

It has been demonstrated in Pasteur's laboratory that glycerin has no appreciable antiseptic influence upon the virus of hydrophobia; therefore, I mixed the virus of hydrophobia with glycerin, and at the expiration of several weeks all the animals which I inoculated with this mixture died with the symptoms of hydrophobia.

On the contrary, when glycerin has been combined with ozone to form glycozone, the compound destroys the hydrophobic virus almost instantaneously.

Two months ago, a rabbit was inoculated with the hydrophobic virus, which had been submitted to the action of this new compound, and the animal is still alive.

I believe that the practitioner will meet with very satisfactory results with the use of peroxide of hydrogen for the following reasons:

1. This chemical seems to have no injurious effect upon animal cells.
2. It has a very energetic destructive action upon vegetable cells—microbes.
3. It has no toxic properties; five cubic centimetres injected beneath the skin of a guinea-pig do not produce any serious result, and it is also harmless when given by the mouth.

As an immediate conclusion resulting from my experiments, my opinion is, that peroxide of hydrogen should be used in the treatment of diseases caused by germs, if the microbial element is directly accessible; and it is particularly useful in the treatment of infectious diseases of the throat and mouth.



Books Received.

Books Received.

THE PHYSICIAN'S VISITING LIST (*Lindsay & Blakiston's*) FOR 1891. Fortieth year of its publication. P. Blakiston, Son & Co., 1012 Walnut street, Philadelphia, Pa.

DESCRIPTIVE ANATOMY OF THE HUMAN TEETH: By G. V. BLACK, M.D., D.D.S. Published by the Wilmington Dental Mfg. Company, 1413 Filbert street, Philadelphia.

"THE PHYSICIAN'S ALL-REQUISITE ACCOUNT BOOK;" being a ledger and account book for physicians' use, meeting all the requirements of the law and courts. Published by F. A. Davis, Medical Publisher and Bookseller, 1231 Filbert street, Philadelphia, Pa. Style No. 1, 900 accounts, price \$5.00, net; style No. 2, 1,800 accounts, \$8.00, net.

A TREATISE ON THE IRREGULARITIES OF THE TEETH AND THEIR CORRECTION; including, with the author's practice, other current methods. Designed for practitioners and students. Illustrated with nearly 2,000 engravings (*not embracing those in the third volume*). Vol. I., by JOHN NUTTING FARRAR, M.D., D.D.S. Herman Helfeld, General Agent, 1271 Broadway, New York. Price of Vol. I., full cloth, \$6.00.

TWELVE LECTURES ON THE STRUCTURE OF THE CENTRAL NERVOUS SYSTEM, for physicians and students, by Dr. LUDWIG EDINGER, Frankfort-on-the-Main. Second revised edition with 133 illustrations. Translated by Willis Hall Vittum, M.D., St. Paul, Minn. Edited by C. Eugene Riggs, A.M., M.D., Professor of Mental and Nervous Diseases, University of Minnesota; Member of the American Neurological Association. F. A. Davis, Publisher, Philadelphia, Pa.

Brief Mention.

Brief Mention.

DELAYED.—This issue has been delayed on account of anticipated changes in this journal, which will be announced in the future.

THE RUSSIAN MEDICAL DEPARTMENT has issued an order that druggists are on no account to dispense medicines on the prescriptions of dentists.

DR. E. M. THOMAS, a student of Dr. Frank Brewer (formerly of Palmyra, Mo.) enjoys a most enviable practice in the beautiful city of Vienna, Austria.

A LADY STUDENT in dentistry is desirous to obtain a situation in a dental office as assistant. Can make herself useful in the laboratory. Address, "Dental Student," 1100 Burlington street, Muscatine, Iowa.

SIR JOHN TOMES is spending his declining years very comfortably and pleasantly at his delightful country home, Upward Gorse, Catram Valley, and devotes a great deal of his time to the literature of the day, and is remarkably strong and well preserved for a man of his age.

THE TOOTH TRANSPLANTED from a bell-boy's mouth to a lady's, the full description of which is given in the *Missouri Dental Journal* of 1882, on page 245, was extracted a few days ago, having done faithful service in its transplanted socket for *ten years*, lacking one month. The roots were considerably absorbed; also the external plate of the socket, which made it unfavorable to implant another tooth, so a "dummy," attached to a gold crown on an adjoining root, was substituted in its place.

WM. N. MORRISON, *St. Louis.*

[END OF VOLUME VII.]

FOOTNOTES.

- [1] Read before Harvard Odontological Society.
- [2] Read before the Fifth District Dental Society of New York State.
- [3] Read before the Southern Dental Society, at Chester, October, 1890.
- [4] Read before the International Medical Congress, held at Berlin, Germany, on the 7th of August, 1890.
- [5] *Gaillard's Medical Journal*, March, 1889.
- [6] The peroxide of hydrogen that I use is manufactured by Mr. Charles Marchand, of New York. This preparation is remarkable for its uniformity in strength, purity and stability.

TRANSCRIBER'S NOTES.

1. Silently corrected simple spelling, grammar, and typographical errors.
 2. Retained anachronistic and non-standard spellings as printed.
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